Chapter 8: Conservation of Mechanical Energy

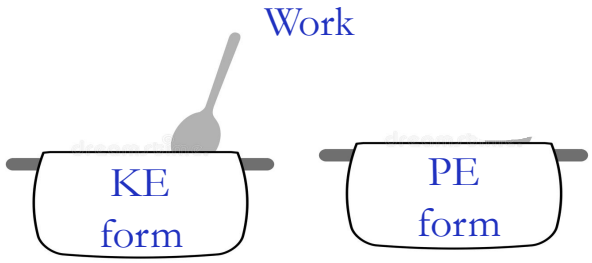
The following subjects will be discussed

- Potential Energy.
- Energy Conservation Law.
- Work by External Forces.
- Examples.

Potential Energy

Energy transforms from one form to another by means of work.

Energy is never "lost".



Example 1: The case of an object thrown upward. The kinetic energy is gradually transformed to gravitational Potential energy. $U = -W_g$

Example 2: The case of a moving object hits and squeezes a spring. The kinetic energy is gradually transformed to spring Potential energy. $U = -W_s$

Example 3: The case of a moving object sliding on a rough surface. The kinetic energy is gradually transformed to Thermal energy (not a potential energy).

Other Examples of potential energy: electrostatic energy. Energy stored in a flexible plastic rod.

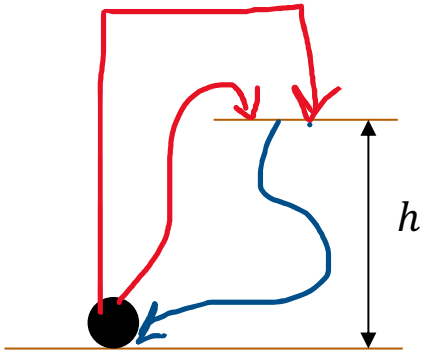
Potential Energy

Potential Energy is "stored". We can retrieve it when needed. It is a function of displacement (position).

Forces that make potential energy are called "conservative" forces.

For conservative forces, work is independent of path.

For conservative forces, work in a closed path is zero.



In one dimension, the change in potential energy is

$$\Delta U = -W = -\int_{x_i}^{x_f} F(x)\,dx$$

For gravitational force, the change in potential energy is

$$\Delta U = mg\,\Delta y \quad \rightarrow \quad U = U_i + mg\,(y - y_i)$$

There should always be a reference, for example $y = 0 \rightarrow U(0) = 0$

For spring force, the change in potential energy is

$$\Delta U = \frac{1}{2}k(x_f^2 - x_i^2)$$

There should always be a reference, for example $x = 0 \rightarrow U(0) = 0$

Conservation of Mechanical Energy

For an isolated system in the presence of conservative forces,

$$\Delta KE + \Delta U = 0$$

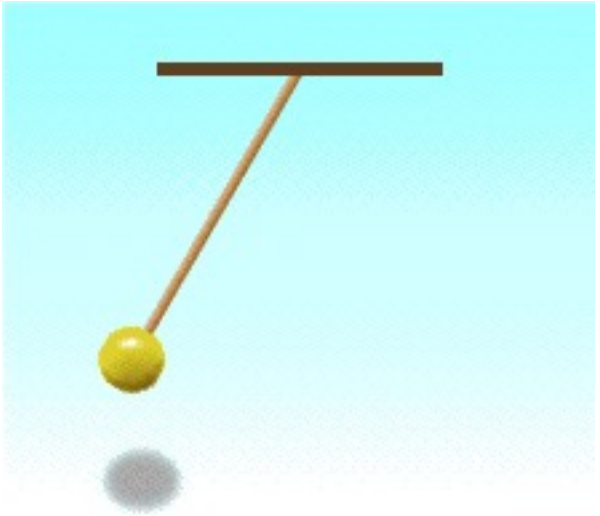
Can be written as,

$$KE_2 - KE_1 = -(U_2 - U_1)$$

Can be written as,

$$KE_1 + U_1 = KE_2 + U_2 = constant$$

Total Energy is
always the same
conserved

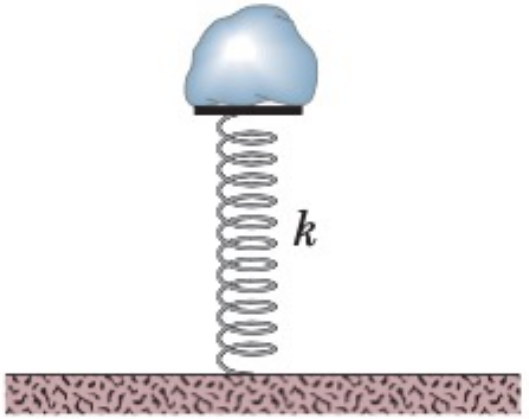


Examples

1) The 8-kg object compressed the spring 10 cm. Later, the object is pushed down further 30 cm. Find the total spring potential energy.

$$U_s = \frac{1}{2}k\left(x_f^2 - x_i^2\right) = \frac{1}{2}k(0.1 + 0.3)^2$$

$$U_s = \frac{1}{2}(784)(0.4)^2 = 62.72\ J$$



When released, who high in the air it will go?

$$U_s \rightarrow U_g = mgh = 62.72\ J$$

$$h = \frac{62.72}{8(9.81)} = 79\ cm$$

$$k = \frac{mg}{x} = \frac{8(9.81)}{0.1} = 784\ N/m$$

Examples

2) The 2-kg object is pushed 20 cm against the spring. $D = 1$ meter, is the distance from the relaxed position of the sprint to the ground level. The object is then released. $k = 170N/m$. The surface is frictionless.

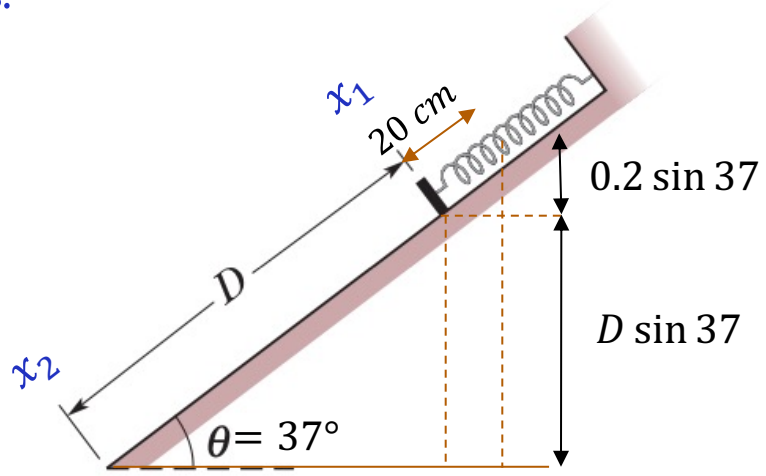
Find the speed of the object at the relaxed position of the spring x_1.

$$U_{tot_x1} = U_s + U_g = \frac{1}{2}kx^2 + mgh$$

$$U_{tot_x1} = \frac{1}{2}(170)(0.2)^2 + 2(9.81)(0.2\sin 37)$$

$$U_{tot_x1} = 5.76\,J = KE_{x1} = \frac{1}{2}mv_1^2$$

$$v_1 = \sqrt{\frac{2(5.76)}{2}} = 2.4\,m/s$$



Find the speed of the object at x_2.

$$KE_2 + U_2 = KE_1 + U_1$$

$$\frac{1}{2}mv_2^2 = \frac{1}{2}mv_1^2 + mg\,D\sin 37$$

$$v_2 = 4.19\,m/s$$

Examples

3) The Pendulum is released from the position shown below with zero initial velocity.

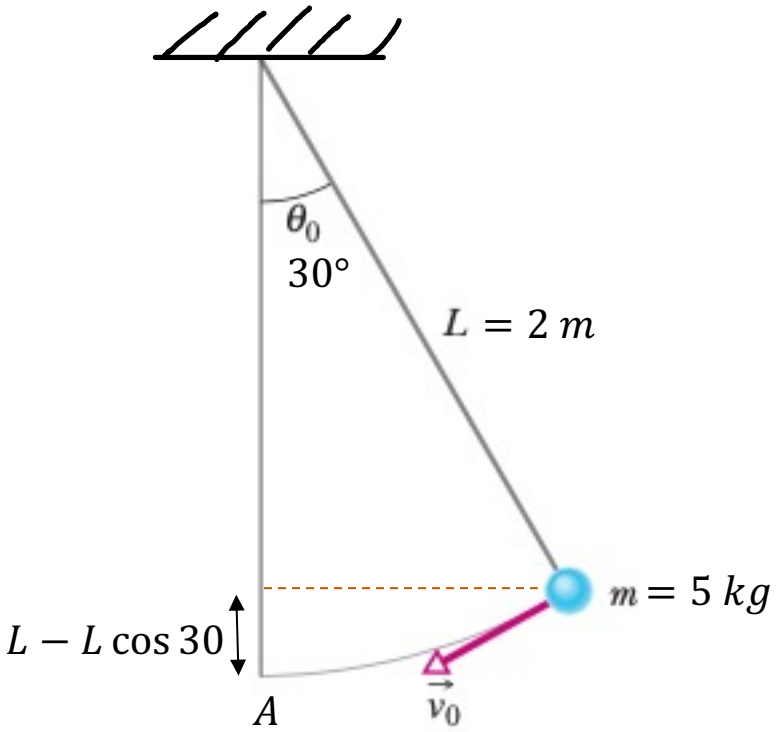
How much work is done by gravitational force when the bob is at the lowest point A?

$$W_g = -\Delta U = mg(L - L\cos 30)$$

$$W_g = 5(9.81)(2 - 2\cos 30) = 13.1\ J$$

How much is the change in potential energy in this case?

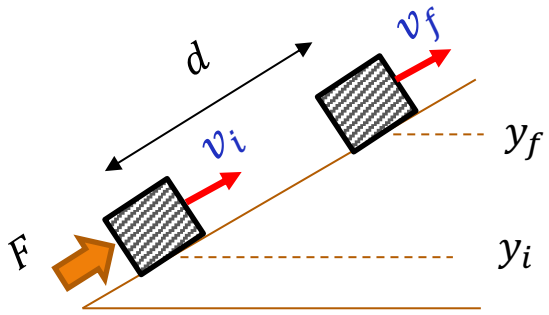
$$\Delta U = -13.1\ J$$



Work Done by External Forces

We expand our discussion of work to include external forces such *F* or friction.

Example: A force pushing an object over a frictionless surface.



$$\Delta KE + \Delta U = W_F = F \cdot d$$

$$\frac{1}{2}m\left(v_f^2 - v_i^2\right) + mg\left(y_f - y_i\right) = F \cdot d$$

For a rough surface

$$\Delta KE + \Delta U = W_{tot} = W_F + W_{friction}$$

$$\Delta KE + \Delta U = F \cdot d - f_k \cdot d$$

Conservation of Energy Principle

For an isolated system, $\Delta E_{mechanical} + \Delta E_{thermal} + \Delta E_{internal} = 0$

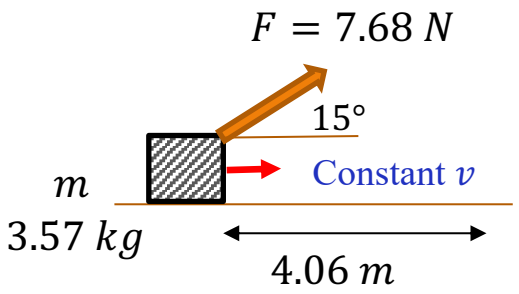
For a non-isolated system, $\Delta E_{mechanical} + \Delta E_{thermal} + \Delta E_{internal} = W$

Power $P = \frac{dE}{dt}$ $P = \frac{\Delta E}{\Delta t}$ Average

Examples

1) The block is moved over a rough surface at a constant speed. Find the work done by the external force F.

$$W_F = F \cdot d = (7.68)\cos 15\ (4.06) = 30.12\ J$$



How much is the kinetic coefficient of friction?

$$\Delta KE + \Delta U = W_{tot} = W_F + W_{friction}$$

$$0 + 0 = 30.12 - \mu_k\ (mg - F\ sin15)\ d$$

$$\mu_k = \frac{30.12}{[(3.57)(9.81) - 7.68\sin 15]\ (4.06)} = 0.21$$

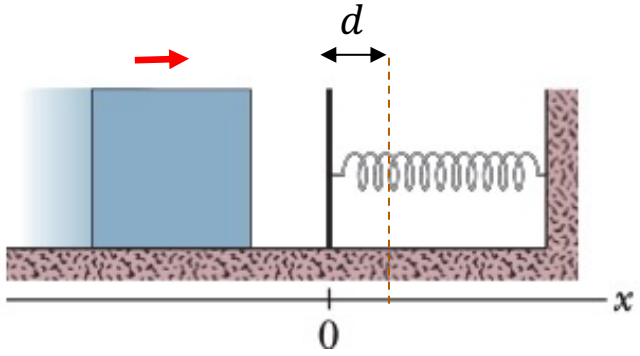
Examples

2) The moving block compresses the spring 7.5 cm as it stops.

How much work is done by the spring force in the process?

$$W_s = -U_s = -\frac{1}{2}k(x_f^2 - x_i^2)$$

$$W_s = -\frac{1}{2}(320)\left(0.075^2 - 0\right) = -0.9\,J$$



$\mu_k = 0.25$

$k = 320\ N/m$

$m = 2.5\ kg$

How much is the increase in the thermal energy of the block-surface system by friction?

$$\Delta E_{friction} = F_{friction} \cdot d = \mu_k mg \cdot d$$

$$\Delta E_{thermal} = (0.25)(2.5)(9.81)(0.075) = 0.46\,J$$

What is the block's speed as it touches the spring?

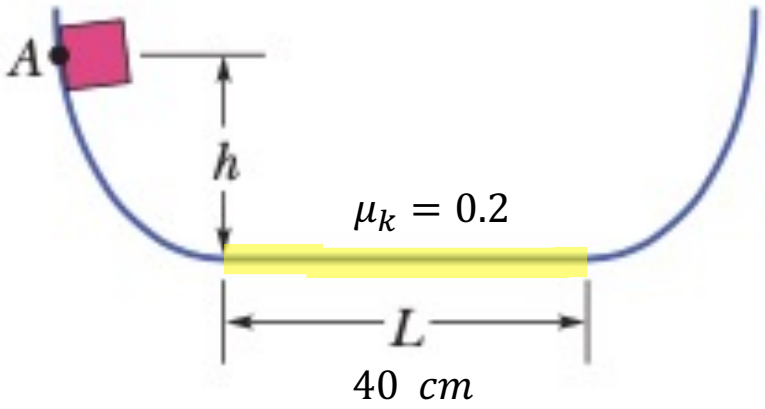
$$\Delta E_{mechanical} + \Delta E_{thermal} + \Delta E_{internal} = W$$

$$\frac{1}{2}mv^2 = 0.9 + 0.46$$

$$v = 1.043\ m/s$$

Examples

3) The shaded part of the track is rough; the rest is frictionless. $h = L/2$.



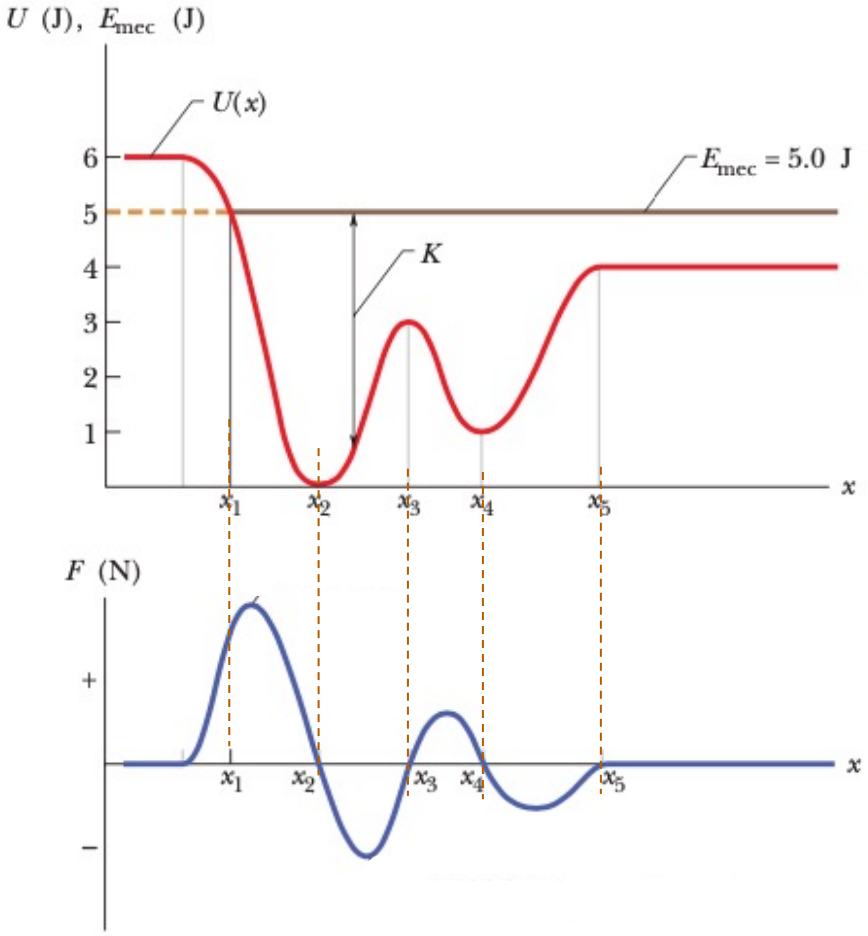
Where will the box finally stop?

Examples

4) The energy curve of a particle moving along the x axis on a frictionless surface due to a conservative force acting on it.

$$U = -\int_{x_i}^{x_f} F(x)\,dx$$

$$UF = -\frac{dU}{dx}$$



$$KE + U = const. = 5\,J$$

Q & A

Send me an email if
you have any
questions.